atrekos

Assessment report: Substation Cellular Router (TW800 Series)

ACME Security · Revision 1 · Completed
Completed 26.10.2026 · Generated 26.10.2026

Industrial 4G LTE cellular router designed for energy sector substation automation. Supports all LTE bands including 450 MHz (B31/B72), Dual SIM with auto-fallback, TPM 2.0, and IPv4/IPv6 dual-stack. Features OpenVPN and IPSec VPN, dual APN for traffic separation, and integrated firewall. DIN-rail metal housing, 12–48 V DC input, high EMC immunity per EN 61000-4. Managed centrally via Smart EMS with zero-touch provisioning.

Residual risk: 10 Very High, 12 High and 90 Moderate of 160 risks, as reported in the companion Result Summary.

The companion Result Summary carries the management brief for this assessment.

CONTENTS

1 Priority findings
2 Security assumptions, constraints, and requirements
3 Scope and confidence
Appendix A Risk register
Appendix B Damage scenarios and questionnaire
Appendix C Control and condition mappings
Appendix D Confirmation records
Appendix E Provenance and traceability
Appendix F Attribution and licences

1 Priority findings

Ordered by residual risk, then by operational consequence, then by open decisions. **Residual risk** is the calculated residual rating for the recorded assessment inputs; treatment and implementation status are shown separately. **Required attack potential (RAP)** is a separate measure of attacker capability, shown under Supporting.

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F1 R8 **Disrupting the product's communication over the wireless network**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate wireless-outage consequences through a proposed operator-owned deployment safeguard plan; no agreed risk transfer is claimed. Remove A29's unvalidated damage exclusions. Define maximum tolerable outage, loss-of-link alarms, safe local behavior, genuinely independent backup or local safeguards, and recovery procedures. Test sustained radio/carrier loss and obtain deployment-owner agreement before commissioning. Dual SIM does not establish independence from a common radio outage. The Basic replacement plan claims no RAP or damage-reduction credit; manufacturer tracking continues. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned operator safeguards for wireless outages - agreement and verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			
F2 R12-4 **Interrupting the product with a denial of service attack**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate product denial of service from untrusted local devices through a planned DoS/resource-isolation response. Specify per-interface resource quotas, essential-function priorities, alarms and recovery, and test floods on the local interface. Remove C20's unvalidated damage transformation; validate operational and safety consequences at system level. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F3 R12-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the untrusted local-device path to unpatched-vulnerability exploitation; no agreed third-party risk transfer is claimed. Define manufacturer fix/support duties, operator release monitoring, patch deadlines, postponement restrictions and integrator advisory delivery. Record a receiving party and agreed handover if transfer is proposed later; retain manufacturer tracking meanwhile and verify update availability and each installation notification route. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	None recorded			
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → Enhanced-Basic			
F4 R13-4 **Interrupting the product with a denial of service attack**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate product denial of service from the untrusted LAN through a planned DoS/resource-isolation response. Define priority traffic, resource quotas, degraded operation and recovery; test representative LAN floods while observing essential communication. Remove C20's unvalidated damage transformation; retain operational/safety consequences pending system-level evidence. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			
F5 R13-14 **Disrupting the product's communication over the local network**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate local-link disruption through a planned local-network-continuity response. Distinguish router resource exhaustion from cable, switch or shared-network failure; define upstream resilience, alarms, independent alternatives and safe local behavior. Test representative link-loss and recovery scenarios. Remove C20's unvalidated damage transformation. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned local-network continuity and independent fallback - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F6 R22-24 **Data extraction from the product's persistent memory**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS2 Disclosure of Personal data (significant)			
Rationale	“Mitigate customer-access persistent-data extraction through planned storage confidentiality/protected keys and IP/extraction protection. Remove C36 as the memory-extraction response and remove A39's customer-trust exclusion. Inventory credentials, keys, configuration, logs, firmware/applications and stored copies; define encryption/key separation, retention limits, maintenance/readout restrictions and offline-extraction tests. Include customer insiders, service, ownership changes and resale. No TPM or other hardware capability is presumed. The Basic replacement plans claim no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned IP extraction protection and distribution controls - verification pending (custom) · Planned persistent-storage confidentiality and protected keys - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			
F7 R27-4 **Interrupting the product with a denial of service attack**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate product denial of service through a planned DoS/resource-isolation response across the accessible communication medium. Specify minimum essential throughput, control latency, resource limits, fallback and recovery; test representative flooding and deployment-level consequences. Remove C20's unvalidated replacement of operational/safety damage with quality-only damage. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F8 R27-14 **Disrupting the product's communication over the local network**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate local-link disruption through a planned local-network-continuity response. Assess media protection, alarms, genuinely independent alternatives, maximum tolerable outage and safe local operation. Test link loss and recovery at the installation level. Remove C20's unvalidated damage transformation; component flooding defenses alone do not establish a reduced consequence. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned local-network continuity and independent fallback - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			
F9 R33 **Disclosure of data by an untrusted entity receiving data from the product**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate disclosure by external recipients through a new planned confidential-data-flow inventory and recipient-control response. Remove A25's no-confidential-data exclusion. Inventory outgoing sensitive payloads, copies and recipients; minimize disclosure, specify recipient authorization/trust, purpose/retention limits and any contractual requirements, and test intended data flows. Encryption in transit does not prevent a legitimate recipient from disclosing plaintext. No recipient agreement or implemented protection is presumed. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned confidential-data flow inventory and recipient controls - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F10 R35 **Disrupting the product's communication with an external entity (with communication over the Internet or a mobile network)**	Very High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate external-connectivity loss through a new planned continuity and safe-local-operation response. Remove A41's quality-only exclusion. Define essential communication, maximum tolerable outage, link-loss alarms, local safe behavior, genuinely independent backup or local safeguards, and operator recovery responsibilities. Test sustained loss of carrier, WAN and required external services, including recovery, with the deployment owner. No fail-safe architecture or operator agreement is presumed. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned external-link continuity and safe local operation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Basic			
F11 R9 **Attacking the product's internal gateway from the Internet**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS15 Major disturbance caused by attacks on customer's assets by manipulated product			
Rationale	“Mitigate the IP-network exposure path with planned device firewall. Define minimal exposed services, restricted management sources, segmentation and patchable service versions; require an external exposure review and exploitation tests. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM1 SCR 2.1 - Device Firewall			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F12 R9-4 **Interrupting the product with a denial of service attack**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate product denial of service through the retained firewall plan and a new planned DoS/resource-isolation response. Specify gateway capacity, permitted-flow stress limits, essential-traffic priorities, degraded operation and recovery. Remove C20's unvalidated conversion of operational/safety damage to quality-only damage; test the actual deployment consequences separately from packet filtering. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM1 SCR 2.1 - Device Firewall · Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → High			
F13 R9-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the IP-network exposure path to unpatched-vulnerability exploitation; no agreed third-party risk transfer is claimed. Retain planned device firewall. Define manufacturer fix/support duties, operator release monitoring, patch deadlines, postponement restrictions and integrator advisory delivery. Record a receiving party and agreed handover if transfer is proposed later; retain manufacturer tracking meanwhile and verify update availability and each installation notification route. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM1 SCR 2.1 - Device Firewall			
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F14 R13-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the untrusted LAN path to unpatched-vulnerability exploitation; no agreed third-party risk transfer is claimed. Define manufacturer fix/support duties, operator release monitoring, patch deadlines, postponement restrictions and integrator advisory delivery. Record a receiving party and agreed handover if transfer is proposed later; retain manufacturer tracking meanwhile and verify update availability and each installation notification route. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	None recorded			
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Moderate → Moderate			
F15 R15-4 **Interrupting the product with a denial of service attack**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate product denial of service from Internet/mobile exposure through the retained firewall plan and a new planned DoS/resource-isolation response. Define ingress capacity, essential communication under flood, degraded operation, alarms and recovery. Validate genuinely independent WAN alternatives and site loss-of-communication behavior. Remove C20's unvalidated damage transformation. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM1 SCR 2.1 - Device Firewall · Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F16 R15-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the Internet/mobile-network path to unpatched-vulnerability exploitation with planned device firewall, operator patch-process requirements, integrator advisory-delivery requirements. Set a critical-patch timeframe and compensating exposure restrictions during postponement; assign each installation an advisory recipient and validate authenticated update availability. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM1 SCR 2.1 - Device Firewall			
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → High			
F17 R17 **Sending manipulated or malicious data to the product (from an untrusted source)**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate the untrusted-input path with planned input validation. Map validation and resource limits to each enabled WebUI, network, SMS, serial and application input; combine fuzzing with semantic authorization and bounds tests. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	C15 CR 3.5 – Input validation			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → Moderate			
F18 R17-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the untrusted-input path to unpatched-vulnerability exploitation; no agreed third-party risk transfer is claimed. Retain planned input validation. Define manufacturer fix/support duties, operator release monitoring, patch deadlines, postponement restrictions and integrator advisory delivery. Record a receiving party and agreed handover if transfer is proposed later; retain manufacturer tracking meanwhile and verify update availability and each installation notification route. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	C15 CR 3.5 – Input validation			
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → Moderate			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F19 R21-24 **Data extraction from the product's persistent memory**	High	**Mitigate**	Alice	30.10.2026

Consequence	DS2 Disclosure of Personal data (significant)
Rationale	“Mitigate third-party persistent-memory extraction through the retained physical-access installation condition and a new planned storage-confidentiality/protected-key response. C36 has been removed as this attack response. Inventory credentials, keys, configuration, logs, code and stored copies; define encryption/key separation, retention limits, raw-read/debug restrictions and offline-extraction tests. Validate the physical-access condition separately. No TPM or other hardware protection is presumed. The Basic replacement plan claims no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	Planned persistent-storage confidentiality and protected keys - verification pending (custom)
Conditions	A14 SDR 1.12 – Product inaccessible
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk Very High · RAP Basic → Moderate

F20 R27-40 **Abusing the lack of a software update functionality to exploit an unpatched vulnerability**	High	**Mitigate**	Alice	30.10.2026

Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors
Rationale	“Mitigate the accessible communication-medium path to unpatched-vulnerability exploitation with planned operator patch-process requirements, integrator advisory-delivery requirements. Define interface-specific compensating restrictions, deadlines and escalation for deferred patches; align the operator/integrator handover with an evidenced maintenance and end-of-support plan. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	None recorded
Conditions	A46 SDR 3.1 - Security updates are installed under the operator’s patch management · A47 SDR 3.2 - Update notifications reach the operator through the integrator
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk High · RAP Moderate → Moderate

F21 R31 **Attacking a customer network from another customer network via the product's internal gateway**	High	**Mitigate**	Alice	30.10.2026

Consequence	DS15 Major disturbance caused by attacks on customer's assets by manipulated product
Rationale	“Mitigate the cross-customer gateway path with planned device firewall. Define trust zones and minimum allowed cross-network flows, test every routed/bridged and tunnel path, and require a verified secure baseline with controlled rule changes before deployment. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	CM1 SCR 2.1 - Device Firewall
Conditions	None recorded
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk Very High · RAP Basic → High

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F22 R22-24-32 **Extracting or misusing ACME Security's IP**	High	**Mitigate**	Alice	30.10.2026
Consequence	DS3 Disclosure of the organisation's IP (slightly innovative)			
Rationale	“Mitigate customer-access persistent-memory IP extraction through planned storage confidentiality/protected keys and IP/extraction protection. Remove C36 as this attack response and remove A39's customer-trust exclusion. Identify proprietary firmware/applications and sensitive copies, confirm IP value, define key separation and readout/maintenance restrictions, and test offline extraction. Include hostile customers, service, ownership changes and resale; distribution restrictions alone are insufficient. The Basic replacement plans claim no RAP or damage-reduction credit. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	Planned IP extraction protection and distribution controls - verification pending (custom) · Planned persistent-storage confidentiality and protected keys - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Basic → Basic			
F23 R2-3 **Compromising the product using a software vulnerability**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the Wi-Fi entry path to software compromise with planned secure development and vulnerability management, secure Wi-Fi configuration. Specify enabled Wi-Fi modes, replace open/default settings, and define vulnerability-management and exploit-test gates before the next revision. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM2 CRA-compliant SDL and vulnerability management · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Moderate → Beyond High			
F24 R2-3-10 **Attacking devices in the customer's network**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS14 Major disturbance caused by manipulated product			
Rationale	“Mitigate the Wi-Fi entry path to customer-device attack with planned secure development and vulnerability management, secure Wi-Fi configuration. Retain the mitigation plan and specify forwarding allowlists, customer-device separation, and pivot tests from a compromised router. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM2 CRA-compliant SDL and vulnerability management · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Moderate → Beyond High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F25 R2-3-11 **Denial of Service on the customer's network**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS14 Major disturbance caused by manipulated product			
Rationale	“Mitigate the Wi-Fi entry path to customer-network denial of service with planned secure development and vulnerability management, secure Wi-Fi configuration. Add design acceptance criteria for egress policing and customer-network flood containment; verify the recalculated residual against documented criteria after those tests. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM2 CRA-compliant SDL and vulnerability management · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Moderate → Beyond High			
F26 R2-4 **Interrupting the product with a denial of service attack**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS7 Major disturbance caused by unavailable product			
Rationale	“Mitigate Wi-Fi-originated product denial of service with the retained secure-Wi-Fi plan and a new planned DoS/resource-isolation response. Define enabled Wi-Fi modes, priority traffic, capacity limits, degraded operation, alarms and recovery; test floods and resource exhaustion at the deployment level. Remove C20's unvalidated damage transformation. The Basic replacement plan claims no RAP or damage-reduction credit; this is a planned mitigation decision, not acceptance of the former automatic Low result. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM20 SCR 2.11 - Secure Wi-Fi · Planned denial-of-service resilience and resource isolation - verification pending (custom)			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Beyond High			
F27 R2-5 **Eavesdropping on the product's communication**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors			
Rationale	“Mitigate the Wi-Fi entry path to interception with planned authenticated encrypted communications, secure Wi-Fi configuration. Identify all traffic requiring confidentiality, require authenticated VPN/TLS protection on those flows, and test downgrade or plaintext fallback. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM11 SCR 2.9 - Secure channel with communication partner authentication · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Basic → Beyond High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F28 R2-6 **Spoofing / tampering with the product's outgoing communication**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS14 Major disturbance caused by manipulated product			
Rationale	“Mitigate the Wi-Fi entry path to outgoing-traffic manipulation with planned authenticated outgoing channels, secure Wi-Fi configuration. Specify endpoint identities, protected command/telemetry paths, key provisioning and replay defenses; test receiver rejection of altered or impostor traffic. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM10 SCR 2.6 - Secure channel with product authentication · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → Beyond High			
F29 R2-7 **Spoofing / tampering with the product's incoming communication**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS11 Personal data (significant) disclosed by manipulated product			
Rationale	“Mitigate the Wi-Fi entry path to incoming-traffic manipulation with planned authenticated encrypted communications, secure Wi-Fi configuration. Define which incoming flows require authenticated channels, reject unauthenticated alternatives, and verify authorization and message validation after authentication. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM11 SCR 2.9 - Secure channel with communication partner authentication · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk Very High · RAP Enhanced-Basic → Beyond High			
F30 R2-7-19-18 **Abusing the software update functionality to manipulate the product**	Moderate	**Mitigate**	Alice	30.10.2026
Consequence	DS11 Personal data (significant) disclosed by manipulated product			
Rationale	“Mitigate the Wi-Fi entry path to malicious update installation with planned authenticated encrypted communications, on-device update authenticity checks, secure Wi-Fi configuration. Specify signed packages, device-side trust-anchor verification, invalid-package rejection, rollback policy and signing-key recovery; test the full update path. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”			
Next action	Implement and verify			
Controls	CM11 SCR 2.9 - Secure channel with communication partner authentication · CM12 Secure update: RE1 Update authenticity and integrity EDR 3.10 / HDR 3.10 / NDR 3.10 – Support for updates · CM20 SCR 2.11 - Secure Wi-Fi			
Conditions	None recorded			
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded			
Supporting	Initial risk High · RAP Moderate → Beyond High			

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F31 R2-7-19-18-10 **Attacking devices in the customer's network**	Moderate	**Mitigate**	Alice	30.10.2026

Consequence	DS14 Major disturbance caused by manipulated product
Rationale	“Mitigate the Wi-Fi entry path to customer-device attack with planned authenticated encrypted communications, on-device update authenticity checks, secure Wi-Fi configuration. Verify update authenticity first, then test customer-network segmentation and least-privilege forwarding following a simulated malicious update. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	CM11 SCR 2.9 - Secure channel with communication partner authentication · CM12 Secure update: RE1 Update authenticity and integrity EDR 3.10 / HDR 3.10 / NDR 3.10 – Support for updates · CM20 SCR 2.11 - Secure Wi-Fi
Conditions	None recorded
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk High · RAP Moderate → Beyond High

F32 R2-7-19-18-11 **Denial of Service on the customer's network**	Moderate	**Mitigate**	Alice	30.10.2026

Consequence	DS14 Major disturbance caused by manipulated product
Rationale	“Mitigate the Wi-Fi entry path to customer-network denial of service with planned authenticated encrypted communications, on-device update authenticity checks, secure Wi-Fi configuration. Define signed-update rejection and outbound resource limits; exercise a compromised-router traffic flood against representative customer-network loads. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	CM11 SCR 2.9 - Secure channel with communication partner authentication · CM12 Secure update: RE1 Update authenticity and integrity EDR 3.10 / HDR 3.10 / NDR 3.10 – Support for updates · CM20 SCR 2.11 - Secure Wi-Fi
Conditions	None recorded
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk High · RAP Moderate → Beyond High

F33 R2-16 **Manipulating the product's configuration via its configuration interface**	Moderate	**Mitigate**	Alice	30.10.2026

Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors
Rationale	“Mitigate the Wi-Fi entry path to unauthorized configuration with planned human authentication, machine-certificate authentication, secure Wi-Fi configuration. Specify unique initial credentials, role-based write rights, authenticated machine management and certificate validation; test every configuration route including remote services. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	CM5 SCR 2.3 - Password based authentication (human) · CM15 SCR 2.16 - Authentication with certificates (machine) · CM20 SCR 2.11 - Secure Wi-Fi
Conditions	None recorded
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk Very High · RAP Basic → Beyond High

© Atrekos. All rights reserved.

1.1 Highest residual risks

FINDING	RESIDUAL RISK	TREATMENT	OWNER	REVIEW / DUE
F34 R2-25 **Accessing the product's debug interfaces remotely**	Moderate	**Mitigate**	Alice	30.10.2026

Consequence	DS1 Disclosure of Personal data (significant) recorded by the product's cameras or sensors
Rationale	“Mitigate the Wi-Fi entry path to remote-debug access with planned debug passwords and independent approval factor, secure Wi-Fi configuration. Document all debug endpoints, disable unnecessary remote access, and specify unique credentials plus the required approval factor; test bypass and recovery paths. The current rating is provisional; use the recalculated risk register. Trace implementation and validation in the next revision; release/deployment requires evidence and accountable owner review.”
Next action	Implement and verify
Controls	CM16 SCR 2.14 - Password-based authentication at debug interfaces · CM20 SCR 2.11 - Secure Wi-Fi
Conditions	None recorded
Status	Treatment recorded: Mitigate · Implementation: Not recorded · Evidence: Not recorded
Supporting	Initial risk Very High · RAP Basic → Beyond High

78 more Moderate residual risks are in the Appendix A.

1.2 Other treatment decisions

Every recorded decision other than Accept that is not featured above.

RISK	TREATMENT	RESIDUAL RISK	OWNER	REVIEW / DUE	IMPLEMENTATION	EVIDENCE
R2-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R2-40	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-7	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-7-19-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-7-19-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-7-19-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-16	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-25	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R9-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded

© Atrekos. All rights reserved.

RISK	TREATMENT	RESIDUAL RISK	OWNER	REVIEW / DUE	IMPLEMENTATION	EVIDENCE
R12-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-5	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-7	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-7-19-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-7-19-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-7-19-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-16	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-25	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R12-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-5	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-6	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-7	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-7-19-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-7-19-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-7-19-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-16	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-25	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R13-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-16	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-25	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R15-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded

© Atrekos. All rights reserved.

RISK	TREATMENT	RESIDUAL RISK	OWNER	REVIEW / DUE	IMPLEMENTATION	EVIDENCE
R17-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R17-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R17-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R20-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R20-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R20-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R21-23	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R21-26	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R21-39	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R22-23	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R22-26	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R22-39	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-3	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-3-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-3-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-5	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-6	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-7	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-7-19-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-7-19-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-7-19-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-16	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-25	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R27-29	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R30	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R34-5	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R34-6	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded

© Atrekos. All rights reserved.

RISK	TREATMENT	RESIDUAL RISK	OWNER	REVIEW / DUE	IMPLEMENTATION	EVIDENCE
R36-5	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R36-7	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R36-7-19-18	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R36-7-19-18-10	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R36-7-19-18-11	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R21-24-32	Mitigate	Moderate	Alice	30.10.2026	Not recorded	Not recorded
R45	Mitigate	Moderate	Unassigned	Not recorded	Not recorded	Not recorded
R2-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R9-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R12-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R13-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R20-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R27-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded
R36-7-19-18-32	Mitigate	Low	Alice	30.10.2026	Not recorded	Not recorded

2 Security assumptions, constraints, and requirements

Obligations this assessment relies on others to fulfil. If a condition is not met, the risk ratings that depend on it no longer hold. Linked risk IDs are in the Appendix C.2.

A14 SDR 1.12 – Product inaccessible

Operator · 7 risks

The product is installed in a physically inaccessible place. It is difficult for an attacker to gain physical access to the product.

Verification
Not confirmed

Hand-over
Passed on by the integrator: The operator is responsible to limit access to the product and its physical interfaces taking adequate measures.

Named party
Customer

© Atrekos. All rights reserved.

Operator, other conditions

CODE	OBLIGATION AND WHY IT MATTERS	VERIFICATION / HAND-OVER	RISKS
A46	Security updates are not installed by the product on its own. The operator schedules them under its own patch management, decides when a release is installed, and may postpone one it cannot take yet — which is what a professional or industrial environment requires and what CRA Annex I (c) allows to be arranged elsewhere where automatic installation is not expected (Recital 56). The operator keeps a process that installs security updates within a defined period of their release. Ratings for 7 risks depend on this condition.	Not confirmed Hand-over: Not recorded	7

Integrator

CODE	OBLIGATION AND WHY IT MATTERS	VERIFICATION / HAND-OVER	RISKS
A47	The product does not notify its users of available security updates by itself. The integrator passes the vendor’s advisories and update notifications to the operator of each installation, so that the party who schedules the update learns that one exists. Where the integrator has handed the installation over, the obligation passes with it. Ratings for 7 risks depend on this condition.	Not confirmed Hand-over: Not recorded	7

Manufacturer organisation

CODE	OBLIGATION AND WHY IT MATTERS	VERIFICATION / HAND-OVER	RISKS
A17	ACME Security's products are produced in a secure operational environment. Ratings for 1 risks depend on this condition.	Not confirmed Hand-over: Not recorded	1
A18	ACME Security's suppliers are trustworthy and their products are not manipulated. The suppliers provide updates when needed. Ratings for 1 risks depend on this condition.	Not confirmed Hand-over: Not recorded	1

3 Scope and confidence

3.1 Target of evaluation

Functionality

Industrial cellular router connecting local equipment networks to remote sites through Ethernet and mobile networks. The product provides routing, NAT, configurable firewall rules, VPN tunnels, local and remote administration, logging and diagnostics. The TOE includes the router hardware, embedded firmware, configuration and supplied networking/security functions. Optional Wi-Fi, serial I/O, positioning and uploaded

Intended use

Business-to-business connectivity for field equipment, including the existing assessment scenario of secondary substations and distributed energy assets. The router transports monitoring and control traffic between local IEDs/RTUs and remote control or management systems, with configurable VPN protection. Installation and maintenance are performed by trained technicians and network administrators. External equipment, carrier

© Atrekos. All rights reserved.

applications depend on hardware variant and configuration.

Foreseeable misuse

Using the router or optional I/O as a safety controller, protection relay or emergency-stop device; using the cellular/VPN link for control that requires guaranteed real-time response or uninterrupted safety-critical connectivity; installing the bare device in exposed outdoor, wet or condensing environments, or operating it outside its specified electrical and temperature limits. These are foreseeable uses outside the defined industrial communications role and operating conditions. Suitability for safety functions or deterministic real-time control is not established.

Network zone

Gateway between operational-equipment LANs and cellular/Ethernet WANs, including potentially Internet-facing connectivity. VPN endpoints connect remote control networks. Management/payload separation and service restrictions depend on configuration; deployed segmentation and peer trust are unconfirmed.

Operator profile

Trained installers, network/security administrators and operational maintenance staff. Administrator accounts have full configuration rights; standard users have read-only monitoring access. External AAA supports RADIUS, TACACS+ and LDAP. Administrators manage credentials, VPN keys/certificates, firewall/service ACLs, backups and firmware updates. Connected equipment exchanges operational traffic.

infrastructure, remote servers and management platforms are outside the TOE boundary.

Deployment context

Industrial embedded appliance installed on a DIN rail or workstation in a field equipment cabinet, including the existing substation/energy deployment scenario. Powered from 12-48 VDC; operating range -25 to +70 C with noncondensing humidity. Customer equipment, carrier networks, remote control/management services and power supply form the operational environment.

Physical access

Normally installed in a field equipment cabinet or controlled technical area accessible to authorized installers and maintenance staff. Site locks and access controls are external assumptions. Hardware is reachable during servicing; reset, SIM, antenna, Ethernet, power and variant-specific console/I/O connections can be tampered with if physical access is obtained. Public access is not intended.

Notes

Evidence basis: reviewed product documentation. Exact hardware variant, installed firmware, deployed configuration and customer applications are unconfirmed.

TOE boundary: physical router, embedded firmware, configuration and supplied networking/security functions. External IEDs/RTUs, customer LANs, carrier infrastructure, management computers and remote VPN/AAA/syslog/control services are operational-environment dependencies. Optional Wi-Fi, serial I/O, positioning and application development functions depend on variant/configuration.

The existing substation/energy use case is retained. Impact assumptions: loss/manipulation of monitoring/control connectivity can cause Major disturbance and potentially dangerous situations. These are deployment assumptions, not manual

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

findings; actual command authority, redundancy and independent safeguards require confirmation. Fatal injury or equipment destruction is not inferred. Proprietary firmware/application IP is assumed slightly innovative.

Foreseeable camera support includes connected network cameras. Accounts, logs, customer traffic, diagnostic packet captures and potentially identifiable camera images support the significant personal-data scenario. Actual payload sensitivity, quantity and retention are unconfirmed.

Service life and planned support are both assumed to be 5-10 years at the user's instruction. This is not an evidenced manufacturer support commitment. Regular field-update planning, product replacement strategy, secure production/suppliers and vulnerability-handling assurances are not demonstrated by the supplied manuals.

Per user instruction, No answers for unevidenced controls/processes mean 'not demonstrated by the reviewed documentation', not proof that the manufacturer lacks them. This covers SBOM/vulnerability documentation, timely/free security patches, separate security updates, security testing, public advisories/disclosure policy/reporting contact, authentic update delivery, automatic updates and committed support. Undocumented individual confidentiality/SLA/legal obligations are also not assumed.

Documented capabilities: admin/read-only roles and AAA; configurable management-service ACLs; backups with optional password encryption/private-key export; logs/syslog; manual firmware upload; dual SIM/failover; firewall/NAT; IPsec/OpenVPN/certificates; Python SDK/IDE debug; optional serial/I/O; packet capture and diagnostics. Capabilities do not establish secure deployed settings or trusted peers.

Earlier claims of TPM 2.0, 450 MHz/B31/B72, gigabit Ethernet, dual-APN separation, IPv6, zero-touch provisioning and a named fleet-management platform are unverified in these manuals and receive no security credit. Secure boot, signed updates,

© Atrekos. All rights reserved.

MFA, comprehensive encryption at rest, audit-log integrity and cybersecurity certification are not established. Exact product/version identifiers and a published DoC URL remain missing; the manual reproduces a CE declaration.

Contacts

Not recorded

Product lifetime

Not recorded

Support duration

Not recorded

Interfaces

NAME	KIND	DESCRIPTION	EXPOSED TO
Cellular WAN and SMS interface	radio	Cellular modem/SIM/antenna connectivity with variant-dependent 2G/3G/4G capability. SMS can query status and issue configured reboot/connection commands using a permitted-number list. Optional Wi-Fi variants can act as access point or client; exact variant and deployment are unconfirmed.	mobile-network-operator, public-internet, optional-wireless-LAN, nearby-radio-parties
Ethernet LAN/WAN interfaces	network	RJ45 connections for local operational equipment and optional Ethernet WAN uplink. Port count and speed depend on the selected hardware variant. Includes routed, switched and forwarded traffic and exposed management services.	substation-lan, ieds, rtus
VPN Tunnel Endpoints (OpenVPN/IPSec)	network	Logical VPN tunnel interfaces over the cellular WAN link to the DSO control center VPN concentrator	control-center, scada-master, vpn-concentrator
Serial console and optional serial/I/O interfaces	serial	Local serial console is documented. RS232/RS485 data channels, digital input and relay output are variant dependent; optional serial forwarding or SMS relay commands may exchange operational data. Exact installed interfaces are unconfirmed.	local-technician, connected-serial-equipment, optional-digital-I/O
Network administration and monitoring	network	HTTP/HTTPS WebUI, Telnet/SSH CLI, SNMP, external AAA and syslog, subject to configurable service and source ACL restrictions. Firmware/configuration imports, diagnostics and optional Python application/IDE debug functions are exposed through administration. A separate fleet-management channel is unverified.	administrator, management-network, external-AAA-and-syslog-services
DC Power Input (12–48 V)	physical	Wide-range DC power input terminal block, supporting 12 V to 48 V DC for flexible substation power supply integration	power-infrastructure
Physical service and status interfaces	physical	Status LEDs, physical factory-reset control and SIM slots. Local service access can reset configuration or alter subscriber identity; variant-specific serial/I/O connections are documented separately. These interfaces are reachable by persons with physical access to the router.	operator, field-technician

Assets

NAME	CATEGORY	SECURITY PROPERTIES	LINKED DS	LINKED STEPS
Cellular connectivity hardware	hardware	Availability, Operational	–	–

© Atrekos. All rights reserved.

NAME	CATEGORY	SECURITY PROPERTIES	LINKED DS	LINKED STEPS
Dual SIM Controller	hardware	Availability	–	–
Cryptographic key and credential storage	data	Confidentiality, Integrity, Availability	–	–
Embedded router firmware	firmware	Confidentiality, Integrity, Availability, Operational	–	–
VPN Tunnel Engine	function	Confidentiality, Integrity, Availability, Operational	–	–
Traffic routing and firewall policy	function	Confidentiality, Integrity, Availability, Operational	–	–
Administration and update functions	function	Confidentiality, Integrity, Availability, Operational	–	–
Configuration Store	data	Confidentiality, Integrity, Availability	–	–
Operational traffic, logs and diagnostics	data	Confidentiality, Integrity, Availability, Operational, Privacy	–	–

3.2 Evidence and confirmation limits

A complete questionnaire records what the assessor answered. It does not verify that controls are in place. Missing verification details do not mean a control is absent.

LIMITATION	IMPLICATION	DETAIL
Hand-over not recorded	No hand-over to the operator is recorded for 1 of 5 operating conditions that can be passed on (A47).	Section 2
Owners and dates	No accountable owner or review date is recorded for 1 of 121 risks with a decision.	Appendix A

3.3 Quality findings

Automated quality indicators: 0 errors, 2 warnings, 8 informational. 2 of 10 have been accepted.

CODE	SEVERITY	ISSUE	DISPOSITION	OWNER	REVIEW BY
QI-1	Info	All active questions answered	Not accepted	Unassigned	Not recorded
QI-4	Info	No dangling prerequisite references	Not accepted	Unassigned	Not recorded
QI-5	Info	No dangling DS references in DSTs	Not accepted	Unassigned	Not recorded
QI-6	Info	All active DS have answered questions	Not accepted	Unassigned	Not recorded
QI-7	Info	All damage selections complete	Not accepted	Unassigned	Not recorded
QI-8	Info	No attack steps that nothing can activate	Not accepted	Unassigned	Not recorded
QI-9	Info	All risks have treatment decisions	Not accepted	Unassigned	Not recorded

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

CODE	SEVERITY	ISSUE	DISPOSITION	OWNER	REVIEW BY
QI-12	Warning	No confirmed mitigations — tracing not run	Accepted: Still in design phase. Tracing in next revision.	Alice	Not recorded
QI-16	Info	2 QI acceptance(s) recorded	Not accepted	Unassigned	Not recorded
QI-17	Warning	Risk concentration: 22 risk(s) at High or Very High	Accepted: Conditional that mitigations are implemented.	Alice	30.10.2026

Questionnaire completeness	49 of 49 active questions answered. Questionnaire completion does not verify controls.
Documentation checks	1 of 3 open. This does not mean warnings are accepted or controls verified.

3.4 Approval and traceability

Approved by Bob Example (assessor) on 26.10.2026. The approval covers these inputs (input hash 7bda1bf4aa5a0c9e).

Completed	26.10.2026
Approved	Bob Example (assessor), 26.10.2026
Approval rationale	Reviewed the assessor’s answers, mitigations and treatments and the computed results; approved as the example assessment.
Input hash at approval	7bda1bf4aa5a0c9e
Input hash of this report	7bda1bf4aa5a0c9e
Report generated	26.10.2026
Product owner	Alice Example
Pack and engine	Standard catalogue pack · engine 0.5.3

© Atrekos. All rights reserved.

APPENDICES

A Risk register

B Damage scenarios and questionnaire

C Control and condition mappings

D Confirmation records

E Provenance and traceability

F Attribution and licences

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

This register is complete: 160 of 160 risks, one row each. It merges the risk register and the countermeasures and residual-risk table that earlier reports printed separately. Every row is a risk of an active attack path. A dash means the value does not exist for that risk. Codes are resolved in C.1 and the confirmation of each in Appendix D.

Ordered by residual risk, highest first, then by risk ID.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R8	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R12-4	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R12-40	Very High	Very High	Mitigate	Enhanced-Basic → Enhanced-Basic	–	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R13-4	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R13-14	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R22-24	Very High	Very High	Mitigate	Basic → Basic	(custom), (custom)	–	Alice	30.10.2026
R27-4	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R27-14	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R33	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R35	Very High	Very High	Mitigate	Basic → Basic	(custom)	–	Alice	30.10.2026
R9	Very High	High	Mitigate	Basic → High	CM1	–	Alice	30.10.2026
R9-4	Very High	High	Mitigate	Basic → High	CM1, (custom)	–	Alice	30.10.2026
R9-40	Very High	High	Mitigate	Enhanced-Basic → High	CM1	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R13-40	High	High	Mitigate	Moderate → Moderate	–	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R15-4	Very High	High	Mitigate	Basic → High	CM1, (custom)	–	Alice	30.10.2026

© Atrekos. All rights reserved.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R15-40	Very High	High	**Mitigate**	Enhanced-Basic → High	CM1	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R17	Very High	High	**Mitigate**	Enhanced-Basic → Moderate	C15	—	Alice	30.10.2026
R17-40	Very High	High	**Mitigate**	Enhanced-Basic → Moderate	C15	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R21-24	Very High	High	**Mitigate**	Basic → Moderate	(custom)	A14 (Operator)	Alice	30.10.2026
R22-24-32	High	High	**Mitigate**	Basic → Basic	(custom), (custom)	—	Alice	30.10.2026
R27-40	High	High	**Mitigate**	Moderate → Moderate	—	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R31	Very High	High	**Mitigate**	Basic → High	CM1	—	Alice	30.10.2026
R2-3	High	Moderate	**Mitigate**	Moderate → Beyond High	CM2, CM20	—	Alice	30.10.2026
R2-3-10	High	Moderate	**Mitigate**	Moderate → Beyond High	CM2, CM20	—	Alice	30.10.2026
R2-3-11	High	Moderate	**Mitigate**	Moderate → Beyond High	CM2, CM20	—	Alice	30.10.2026
R2-4	Very High	Moderate	**Mitigate**	Basic → Beyond High	CM20, (custom)	—	Alice	30.10.2026
R2-5	Very High	Moderate	**Mitigate**	Basic → Beyond High	CM11, CM20	—	Alice	30.10.2026
R2-6	Very High	Moderate	**Mitigate**	Enhanced-Basic → Beyond High	CM10, CM20	—	Alice	30.10.2026
R2-7	Very High	Moderate	**Mitigate**	Enhanced-Basic → Beyond High	CM11, CM20	—	Alice	30.10.2026
R2-7-19-18	High	Moderate	**Mitigate**	Moderate → Beyond High	CM11, CM12, CM20	—	Alice	30.10.2026
R2-7-19-18	High	Moderate	**Mitigate**	Moderate → Beyond High	CM11, CM12, CM20	—	Alice	30.10.2026
R2-7-19-18	High	Moderate	**Mitigate**	Moderate → Beyond High	CM11, CM12, CM20	—	Alice	30.10.2026
R2-16	Very High	Moderate	**Mitigate**	Basic → Beyond High	CM5, CM15, CM20	—	Alice	30.10.2026

© Atrekos. All rights reserved.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R2-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM16, CM20	–	Alice	30.10.2026
R2-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32, CM20	–	Alice	30.10.2026
R2-40	High	Moderate	Mitigate	Moderate → Beyond High	CM20	A46 (Operator), A47 (Integrator)	Alice	30.10.2026
R9-3	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R9-3-10	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R9-3-11	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R9-7	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM11	–	Alice	30.10.2026
R9-7-19-18	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM11, CM12	–	Alice	30.10.2026
R9-7-19-18	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM11, CM12	–	Alice	30.10.2026
R9-7-19-18	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM11, CM12	–	Alice	30.10.2026
R9-16	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM1, CM5, CM15	–	Alice	30.10.2026
R9-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM1, CM16	–	Alice	30.10.2026
R9-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32, CM1	–	Alice	30.10.2026
R12-3	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM2	–	Alice	30.10.2026
R12-3-10	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM2	–	Alice	30.10.2026
R12-3-11	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM2	–	Alice	30.10.2026
R12-5	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM11	–	Alice	30.10.2026
R12-7	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11	–	Alice	30.10.2026
R12-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R12-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R12-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R12-16	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM5, CM15	–	Alice	30.10.2026
R12-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM16	–	Alice	30.10.2026
R12-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32	–	Alice	30.10.2026
R13-3	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R13-3-10	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R13-3-11	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R13-5	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM11	–	Alice	30.10.2026
R13-6	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM10	–	Alice	30.10.2026
R13-7	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11	–	Alice	30.10.2026
R13-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R13-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R13-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R13-16	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM5, CM15	–	Alice	30.10.2026
R13-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM16	–	Alice	30.10.2026
R13-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32	–	Alice	30.10.2026
R15-3	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R15-3-10	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R15-3-11	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM1, CM2	–	Alice	30.10.2026
R15-16	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM1, CM5, CM15	–	Alice	30.10.2026
R15-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM1, CM16	–	Alice	30.10.2026
R15-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32, CM1	–	Alice	30.10.2026
R17-3	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	C15, CM2	–	Alice	30.10.2026

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R17-3-10	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	C15, CM2	–	Alice	30.10.2026
R17-3-11	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	C15, CM2	–	Alice	30.10.2026
R20-18	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM12	–	Alice	30.10.2026
R20-18-10	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM12	–	Alice	30.10.2026
R20-18-11	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM12	–	Alice	30.10.2026
R21-23	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM7	A14 (Operator)	Alice	30.10.2026
R21-24-32	High	Moderate	Mitigate	Basic → Moderate	(custom)	A14 (Operator)	Alice	30.10.2026
R21-26	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM8	A14 (Operator)	Alice	30.10.2026
R21-39	High	Moderate	Mitigate	High → Beyond High	CM21	A14 (Operator)	Alice	30.10.2026
R22-23	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM7, (custom)	–	Alice	30.10.2026
R22-26	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM8, (custom)	–	Alice	30.10.2026
R22-39	High	Moderate	Mitigate	High → Beyond High	CM21, (custom)	–	Alice	30.10.2026
R27-3	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R27-3-10	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R27-3-11	High	Moderate	Mitigate	Moderate → Beyond High	CM2	–	Alice	30.10.2026
R27-5	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM11	–	Alice	30.10.2026
R27-6	High	Moderate	Mitigate	Moderate → Beyond High	CM10	–	Alice	30.10.2026
R27-7	High	Moderate	Mitigate	Moderate → Beyond High	CM11	–	Alice	30.10.2026
R27-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R27-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R27-7-19-1	High	Moderate	Mitigate	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R27-16	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM5, CM15	–	Alice	30.10.2026
R27-25	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM16	–	Alice	30.10.2026
R27-29	Very Hi	Moderate	Mitigate	Basic → Beyond High	C32	–	Alice	30.10.2026
R30	Very Hi	Moderate	Mitigate	Basic → Beyond High	C17	–	Alice	30.10.2026
R34-5	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM11	–	Alice	30.10.2026
R34-6	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM10	–	Alice	30.10.2026
R36-5	Very Hi	Moderate	Mitigate	Basic → Beyond High	CM11	–	Alice	30.10.2026
R36-7	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11	–	Alice	30.10.2026
R36-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R36-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R36-7-19-1	Very Hi	Moderate	Mitigate	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R37	Very Hi	Moderate	Accept	Basic → Beyond High	–	A17 (Manufacturer organisation)	Alice	30.10.2026
R38	Very Hi	Moderate	Accept	Basic → Beyond High	–	A18 (Manufacturer organisation)	Alice	30.10.2026
R41	High	Moderate	Accept	High → Beyond High	C17	–	Alice	30.10.2026
R45	High	Moderate	Mitigate	Basic → Moderate	CM39	–	—	—
R2-3-32	Modera	Low	Accept (proposal)	Moderate → Beyond High	CM2, CM20	–	—	—
R2-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11, CM20	–	—	—
R2-7-19-18	Modera	Low	Mitigate	Moderate → Beyond High	CM11, CM12, CM20	–	Alice	30.10.2026

© Atrekos. All rights reserved.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R2-7-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM11, CM20	–	–	–
R2-16-32	High	Low	Accept (proposal)	Basic → Beyond High	CM5, CM15, CM20	–	–	–
R2-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM16, CM20	–	–	–
R2-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32, CM20	–	–	–
R9-3-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM1, CM2	–	–	–
R9-7-19-18	Modera	Low	**Mitigate**	Enhanced-Basic → Beyond High	CM1, CM11, CM12	–	Alice	30.10.2026
R9-7-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM1, CM11	–	–	–
R9-16-32	High	Low	Accept (proposal)	Basic → Beyond High	CM1, CM5, CM15	–	–	–
R9-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM1, CM16	–	–	–
R9-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32, CM1	–	–	–
R12-3-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM2	–	–	–
R12-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11	–	–	–
R12-7-19-1	Modera	Low	**Mitigate**	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R12-7-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM11	–	–	–
R12-16-32	High	Low	Accept (proposal)	Basic → Beyond High	CM5, CM15	–	–	–
R12-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM16	–	–	–
R12-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32	–	–	–
R13-3-32	Modera	Low	Accept (proposal)	Moderate → Beyond High	CM2	–	–	–
R13-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11	–	–	–
R13-7-19-1	Modera	Low	**Mitigate**	Moderate → Beyond High	CM11, CM12	–	Alice	30.10.2026
R13-7-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM11	–	–	–
R13-16-32	High	Low	Accept (proposal)	Basic → Beyond High	CM5, CM15	–	–	–

© Atrekos. All rights reserved.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R13-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM16	—	—	—
R13-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32	—	—	—
R15-3-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM1, CM2	—	—	—
R15-16-32	High	Low	Accept (proposal)	Basic → Beyond High	CM1, CM5, CM15	—	—	—
R15-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM1, CM16	—	—	—
R15-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32, CM1	—	—	—
R17-3-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	C15, CM2	—	—	—
R20-18-32	Modera	Low	**Mitigate**	Enhanced-Basic → Beyond High	CM12	—	Alice	30.10.2026
R21-23-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM7	A14 (Operator)	—	—
R21-26-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM8	A14 (Operator)	—	—
R22-23-32	Modera	Low	**Accept**	Enhanced-Basic → Beyond High	CM7, (custom)	—	Alice	30.10.2026
R22-26-32	Modera	Low	**Accept**	Enhanced-Basic → Beyond High	CM8, (custom)	—	Alice	30.10.2026
R27-3-32	Modera	Low	Accept (proposal)	Moderate → Beyond High	CM2	—	—	—
R27-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11	—	—	—
R27-7-19-1	Modera	Low	**Mitigate**	Moderate → Beyond High	CM11, CM12	—	Alice	30.10.2026
R27-7-32	Modera	Low	Accept (proposal)	Moderate → Beyond High	CM11	—	—	—
R27-16-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM5, CM15	—	—	—
R27-25-32	High	Low	Accept (proposal)	Basic → Beyond High	CM16	—	—	—
R27-29-32	High	Low	Accept (proposal)	Basic → Beyond High	C32	—	—	—
R34-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11	—	—	—
R36-5-32	High	Low	Accept (proposal)	Basic → Beyond High	CM11	—	—	—

© Atrekos. All rights reserved.

A Risk register

RISK	INITIAL	RESIDUAL	TREATMENT	RAP INITIAL → RESIDUAL	CONTROLS	CONDITIONS	OWNER	REVIEW BY
R36-7-19-1	Modera	Low	**Mitigate**	Enhanced-Basic → Beyond High	CM11, CM12	–	Alice	30.10.2026
R36-7-32	Modera	Low	Accept (proposal)	Enhanced-Basic → Beyond High	CM11	–	—	—

B Damage scenarios and questionnaire

B.1 Damage scenarios

This table lists every damage scenario of the pinned catalogue: 30. 14 are active in this assessment. A scenario is active when the answers to its question satisfy its activation rule; the engine decides, and inactive scenarios are listed for completeness.

ID	SCENARIO	DAMAGE	ACTIVE
DS1	Disclosure of Personal data (significant) recorded by the product's cameras or sensors	High (4)	Yes
DS2	Disclosure of Personal data (significant)	High (4)	Yes
DS3	Disclosure of the organisation's IP (slightly innovative)	Low (2)	Yes
DS4	Fines for leaking data of another party	—	No
DS5	Disclosure of cryptographic material causing loss of Major disturbance	High (4)	Yes
DS6	Potentially dangerous situation caused by unavailable product	Low (2)	Yes
DS7	Major disturbance caused by unavailable product	High (4)	Yes
DS8	Loss of caused by unavailable product	—	No
DS9	Contract-related fines of caused by unavailable product	—	No
DS10	Potentially dangerous situation caused by manipulated product	Low (2)	Yes
DS11	Personal data (significant) disclosed by manipulated product	High (4)	Yes
DS12	IP (slightly innovative) disclosed by manipulated product	Low (2)	Yes
DS13	Major disturbance caused by a disclosure or misuse of cryptographic material by the manipulated product	High (4)	Yes
DS14	Major disturbance caused by manipulated product	High (4)	Yes
DS15	Major disturbance caused by attacks on customer's assets by manipulated product	High (4)	Yes
DS16	Fines of caused by manipulated product	—	No
DS17	Disturbance of function until restore (Service required)	—	No
DS18	Limited response of the product during DoS attack, recovery afterwards (Disturbance)	—	No
DS19	ACME Security can proof misuse of patented IP (slightly innovative) in the product	—	No

© Atrekos. All rights reserved.

Input hash 7bda1bf4aa5a0c9e

ID	SCENARIO	DAMAGE	ACTIVE
DS5b	Disclosure of user data material causing loss of	—	No
DS13b	caused by a disclosure or misuse of user data by the manipulated product	—	No
DS20	caused by spoofed message	—	No
DS21	Personal data (low) disclosed by spoofed message or impersonation	Low (2)	Yes
DS22	caused by spoofed message	—	No
DS23	caused by unauthenticated operations	—	No
DS24	caused by unauthorized access/operation at firmware/OS-level	—	No
DS25	caused by unauthorized access/operation at application-level	—	No
DS26	Maintenance and root-cause analysis hindered by insufficient diagnostic data and security logs	—	No
DS26b	Disturbance caused by manipulated logs	Low (2)	Yes
DS27	Fines of caused by the impossibility to trace accountability	—	No

B.2 Questionnaire

Every active question with the answer recorded for it, and any other question whose answer carries a rationale.

CODE	QUESTION	ANSWER	RATIONALE
QI1	Does the product have / support cameras or other sensors that can potentially identify people? (Regarding "supporting a camera / sensor": only answer with Yes if attaching a camera / sensor is in the intended or foreseeable use of the product.)	Yes (No)	—
QI2	Does the product process personal data? Examples: address labels	Yes (Personal data (significant))	The router processes account/network identifiers, logs, credentials and transported customer traffic or diagnostic packet captures. Network cameras are foreseeable connected equipment, so identifiable images may be carried and exposed if the router is compromised. Significant personal-data impact is a conservative scenario assumption; actual payload, sensitivity, volume and retention are unconfirmed.
QI3	Does the product contain ACME Security's or your customers' intellectual property? Note: Only IP which is part of / accessible by the firmware / software in the product is in scope of this question. Gaining access to optical and mechanical IP are not covered by this question (and the cybersecurity risk analysis)	Yes (IP (slightly innovative))	Embedded router firmware, protocol/security implementations and any uploaded application code. Conventional routing/VPN functionality is assumed to have limited innovation value; no high-value customer algorithm or proprietary process logic is evidenced. Exact commercial IP value requires owner confirmation.

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QI4	Is ACME Security (typically) obliged to keep the processed / stored data (including third-party IP) confidential by contract or law? (Note that this concerns individual obligations, not the standard product warranty).	No	—
QI5	Does the product contain credentials / cryptographic keys that allow access to sensitive data or to systems in the product's operational environment? (Note: this includes cryptographic data introduced by the customer.)	Yes (Major disturbance)	VPN keys/certificates and local or external-AAA credentials can grant access to router management and connected operational networks. Affected assets include router configuration, remote VPN/control systems and IED/RTU communications. Major disturbance assumes loss or misuse of this access disrupts field monitoring/control; fatal injury or plant destruction is not established by the manuals.
QI5b	Is the device meant to store user data (e.g., customer's applications and configurations)? Credentials and cryptographic keys, including those the customer introduces, are covered by the question above.	No	—
QI6	Can a loss of availability of the product (or its communication) endanger the health of people? (Consider how the operational environment reacts to a loss of availability of the product.)	Yes (Potentially dangerous situation)	Assessment scenario: interruption of IED/RTU monitoring/control communication can delay detection or response to unsafe field conditions. Potentially dangerous situation is an assumed deployment consequence; the router is not a safety function and independent interlocks, redundancy and fail-safe responses are unconfirmed. No life-threatening consequence is established by the manuals.
QI7	Is the availability of the product relevant for its operational environment?	Yes (Major disturbance)	—
QI8	Can a loss of availability allow an attacker access to assets of the customer (including physical access not detected by ACME Security's product)? Example: The product is used for access control.	No	—
QI9	Can a loss of availability cause contract-related fines (e.g., based on a service level agreement)?	No	—
QI10	Can a manipulated product (or its communication) endanger the health of people? Also consider cases where the product is used against its intended use or misconfigured!	Yes (Potentially dangerous situation)	Assessment scenario: malicious routing, firewall or VPN changes, or altered operational traffic, could prevent correct monitoring/control or contribute to unsafe field operation. The potentially dangerous rating reflects the existing substation use case and foreseeable misconfiguration. Downstream command validation, independent safety safeguards and redundancy are unconfirmed; fatal injury is not established by the manuals.

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QI11	Can a manipulated product disclose personal data to an attacker? Also consider cases where the product is used against its intended use or misconfigured!	Yes (Personal data (significant))	A compromised router may expose account-linked identifiers, logs, packet captures and customer traffic. Network cameras are foreseeable connected equipment and can carry identifiable images. Significant impact assumes such images or other personal payload are accessible following compromise; actual data categories, scale and deployed encryption require confirmation.
QI12	Can a manipulated product disclose intellectual property to an attacker? Also consider cases where the product is used against its intended use or misconfigured! Note that this includes both ACME Security's and your customers' IP.	Yes (IP (slightly innovative))	Embedded router firmware and uploaded application code; potentially customer network/configuration information carried in backups or diagnostics. The rating assumes conventional routing/security logic rather than a demonstrated highly innovative algorithm or substantial proprietary process data. Owner validation is needed for actual IP value.
QI13	Can a manipulated product disclose credentials / cryptographic material to an attacker? Also consider cases where the product does not disclose the material but misuses its access to other systems! (E.g., product contains access credentials to store data on a server.)	Yes (Major disturbance)	VPN certificates/private keys, pre-shared secrets, local administrator passwords and external-service/AAA credentials. Disclosure or misuse could give access to router configuration and operational networks, disrupt routing/VPN links or support further attacks on IED/RTU and remote management/control systems. Major disturbance is the retained deployment assumption; no fatal injury or plant destruction claim is inferred.
QI14	Can a manipulated product affect the operation of its operational environment?	Yes (Major disturbance)	—
QI15	Can a manipulated product allow an attacker access to assets of the customer or allow the attacker to attack the customer's network?	Yes (Major disturbance)	—
QI16	Can a manipulated product cause contract-related / legal fines? (Note that this concerns individual obligations, not the standard product warranty).	No	—
QIE1	Can the product receive and/or send unauthenticated messages/commands (e.g., from unauthenticated protocols) that can endanger the health of people?	No	—
QIE2	Does the product support authentication at any level (e.g., RADIUS, logins, JWTs, session cookies, MAC Authentication Bypass)?	Yes (Personal data (low))	Router management credentials and session tokens (loal or external-AAA logins, RADIUS secrets, web/API sssions) could leak through unauthenticated access. Afected assets are these credentials and the account dentifiers they carry. Low personal-data impact is aconservative scenario assumption; no further persona data is evidenced on this path.
QIE3	Can unauthenticated messages from/to the product affect the operational environment (e.g., from/to a CAN bus)?	No	—

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QIE4	Can the product internals (e.g., firmware/OS, file system) be accessed without authentication? Consider all interfaces and answer "yes" if there is at least one way to gain unauthenticated access to the product.	No	—
QIE5	Can a user or an attacker reach functions at firmware/OS level without an authorisation scheme being enforced by default?	No	—
QIE6	Can a user or an attacker reach functions at application level without an authorisation scheme being enforced by default?	No	—
QIE7	Does the product not maintain any log or would the logged information be insufficient for diagnostic purposes?	No	—
QIE7b	Does the product maintain security logs?	Yes (Disturbance)	Router security logs (authentication, configuration-cange and firewall/VPN events) could be manipulated s they are no longer trustworthy for diagnosis or incdent analysis. Affected assets are the security logsand the event history they provide. Disturbance assues the operator can restore trustworthy logging by rset or reconfiguration; no regulatory traceability rquirement is evidenced (QIE8 = No).
QIE8	Do the operations or data exchanges performed on or by the product require traceability for contractual agreements or regulatory compliance? (e.g., post mortem analysis or forensics)	No	—
QA1	Does the product have any connectivity?	Yes	—
QA2	Does the product connect two networks to each other (e.g., as a gateway)?	Yes (Internet-facing gateway)	—
QA3	Does the product have a point-to-point connection to another device?	Yes (No)	—
QA4	Does the product have a direct connection to a local network?	Yes (No)	—
QA5	Does the product support wireless connections?	Yes (No)	Cellular modem/client connecting over 2G/3G/4G carrier networks; SIM/APN settings and VPN endpoints are configured by administrators. SMS management is also documented. Optional hardware variants provide Wi-Fi access point/client functions. Exact variant and deployed service restrictions are unconfirmed; carrier access control alone does not establish trustworthy communicating peers.
QA6	Does the product have a direct connection to a mobile network / the Internet?	Yes	—

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QA7	Does the product offer services to other entities on the network?	Yes	—
QA8	Does the product offer a configuration interface?	Yes (Yes)	WebUI/CLI configuration requires local administrator credentials or configured external AAA; standard accounts are read-only. Configurable service enablement, source-IP/port ACLs, HTTPS/SSH and inactivity timeouts restrict access. This is a documented mechanism, not confirmation of the deployed settings; default credentials must be changed and insecure/exposed services restricted.
QA9	Does the product have an IP stack (or another routable protocol)?	Yes	—
QA10a	Does the product communicate with an external system locally (e.g., a USB stick, an SD card, an RFID tag)?	Yes (No)	SIM interfaces exchange subscriber/authentication data with the local modem. The serial console exchanges administration commands; variant-dependent RS232/RS485 and digital I/O may exchange field-equipment data or control signals. Optional interfaces and connected devices are not confirmed and are not assumed trustworthy. No USB or SD interface is inferred.
QA10b	Does the product communicate with an external system on the Internet (e.g., an untrusted 3rd party backend, a vendor cloud service, a vendor server system)?	Yes (No)	Remote VPN peers and control/management systems exchange operational traffic, routing information and VPN authentication material. Configured services may communicate with external AAA, DNS/DDNS, NTP, syslog, email and certificate/SCEP servers. Exact deployment and peer trust are unconfirmed, so external systems are treated as untrusted in the worst case.
QA11a	Can the product's software be updated over an interface?	Yes (No)	—
QA11b	Are software updates in the field (regularly or event-based) planned?	No (No)	—
QA12	Does the product have a logical connection to a local network over a gateway?	Yes (No)	—
QA13	Does the product have a logical connection to a wireless network over a gateway?	No	—
QA14	Is the product typically installed in a physically accessible position?	Yes	—
QA15	Is the product typically installed in an operational environment with public access?	No	—
QA16	Are the product's physical connections (such as attached bus cables) typically installed in an environment with public access?	No	—
QA17	Does the product have debug / development interfaces (e.g., JTAG, root access over ssh)?	Yes (Yes)	—

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QA18	Is the product produced in a secure environment?	No	—
QA19	Are the supply chain partners trustworthy and protected against attacks?	No	—
QA20	What is the product's expected lifetime?	5 - 10 years	—
QA21	What is the product's planned support duration?	5 - 10 years	—
QVH1	Are vulnerabilities and components contained in the product identified and documented?	No	—
QVH2	Is a software bill of materials (SBOM) maintained in a commonly used, machine-readable format covering at least the top-level dependencies?	No	—
QVH3	Are vulnerabilities addressed and remediated without delay, including by providing security updates?	No	—
QVH4	Where technically feasible, are security updates provided separately from functionality updates?	No	—
QVH5	Are effective and regular tests and reviews of the product security applied?	No	—
QVH6	Once a security update is available, is information about the fixed vulnerabilities shared and publicly disclosed (description, affected products, impact, severity, remediation)?	No	—
QVH7	Is a policy on coordinated vulnerability disclosure in place, enforced and published?	No	—
QVH8	Is a contact address for reporting vulnerabilities (including in third-party components) provided and published?	No	—
QVH9	Are updates distributed through a secure mechanism that protects their integrity and authenticity?	No	—
QVH10	Where applicable, are security updates installed in an automatic manner (with user notification)?	No	—
QVH11	Are security updates disseminated without delay and free of charge, accompanied by advisory messages?	No	—

© Atrekos. All rights reserved.

CODE	QUESTION	ANSWER	RATIONALE
QVH12	Is a support period committed during which vulnerabilities are handled and security updates are provided?	No	—

C Control and condition mappings

C.1 Codes used

CODE	NAME	KIND	RESPONSIBLE PARTY	RISKS
C15	CR 3.5 – Input validation	Control	—	6
C17	CR 4.2 – Information persistence	Control	—	2
C32	SRS 1.11 - Unsuccessful login attempts	Control	—	12
CM1	SCR 2.1 - Device Firewall	Control	—	32
CM2	CRA-compliant SDL and vulnerability management	Control	—	28
CM5	SCR 2.3 - Password based authentication (human)	Control	—	12
CM7	SCR 2.5 - Secure boot	Control	—	4
CM8	SCR 2.7 - Close debug interfaces	Control	—	4
CM10	SCR 2.6 - Secure channel with product authentication	Control	—	4
CM11	SCR 2.9 - Secure channel with communication partner authentication	Control	—	48
CM12	Secure update: RE1 Update authenticity and integrity EDR 3.10 / HDR 3.10 / NDR 3.10 – Support for updates	Control	—	28
CM15	SCR 2.16 - Authentication with certificates (machine)	Control	—	12
CM16	SCR 2.14 - Password-based authentication at debug interfaces	Control	—	12
CM20	SCR 2.11 - Secure Wi-Fi	Control	—	21
CM21	SCR 2.12 - Trusted Execution Environment (TEE)	Control	—	2
CM39	SCR 2.27 - Continuous security monitoring and integrity alerting	Control	—	1
A14	SDR 1.12 – Product inaccessible	Condition	Operator	7
A17	SDR 2.4 – Secure production	Condition	Manufacturer organisation	1
A18	SDR 2.5 – Secure supply chain	Condition	Manufacturer organisation	1
A46	SDR 3.1 - Security updates are installed under the operator’s patch management	Condition	Operator	7
A47	SDR 3.2 - Update notifications reach the operator through the integrator	Condition	Integrator	7

© Atrekos. All rights reserved.

C.2 Risks affected by each condition

A14 · Operator · 7 risks

R21-23, R21-23-32, R21-24, R21-24-32, R21-26, R21-26-32, R21-39

A17 · Manufacturer organisation · 1 risks

R37

A18 · Manufacturer organisation · 1 risks

R38

A46 · Operator · 7 risks

R2-40, R9-40, R12-40, R13-40, R15-40, R17-40, R27-40

A47 · Integrator · 7 risks

R2-40, R9-40, R12-40, R13-40, R15-40, R17-40, R27-40

D Confirmation records

Planning baseline: no mitigations confirmed yet, 22 planned.

E Provenance and traceability

Computed by engine 0.5.3. Identical answers on the same pinned versions give the same input hash and the same results.

ITEM	VERSION OR VALUE
Engine	0.5.3
Model Version	1.0
Questions Catalog Version	1.0
Damage Scenarios Catalog Version	1.0
Attack Steps Catalog Version	1.0
Assumptions Catalog Version	1.0
Countermeasures Catalog Version	1.0
Cyber Resilience Act (Regulation (EU) 2024/2847): Annex I Part I, Annex I Part II, Annex II	1.0
ISA/IEC 62443-4-2	1.0
Radio Equipment Directive 2014/53/EU, Article 3(3)(d), (e): EN 18031-1, -2	1.0
Input hash (SHA-256, first 64 bits)	7bda1bf4aa5a0c9e
Input hash at approval	7bda1bf4aa5a0c9e
Report generated	26.10.2026

© Atrekos. All rights reserved.

F Attribution and licences

These notices cover the catalogue pack as a whole, including content added in versions later than the ones pinned for this report (Appendix E).

Catalog content of	Standard
Creator	Fraunhofer-Institut für Angewandte und Integrierte Sicherheit AISEC and SICK AG
Copyright	QuBA-libre is Copyright (c) by Fraunhofer-Institut für Angewandte und Integrierte Sicherheit AISEC and SICK AG. All Rights Reserved.
License	CC BY-SA 4.0 <https://creativecommons.org/licenses/by-sa/4.0/>
Source	https://github.com/SICKAG/QuBA-libre
Derived from	"QuBA-libre" <https://github.com/SICKAG/QuBA-libre> (CC BY-SA 4.0)
Modifications	The upstream spreadsheet workbook (version 1.1.2) was imported into relational catalogs (questions, damage scenarios, attack steps, assumptions, countermeasures and standards) for use in this application, and its content was changed. Added: 28 questions (23 asked, 5 headings), 11 damage scenarios, 2 attack steps, 2 assumptions and 6 countermeasures; 18 attack steps read the added questions and two of them gain a trigger; 16 countermeasure-to-step links. Rewritten: 51 countermeasure descriptions (38 upstream references replaced by authored text, 13 upstream texts extended or reworded); 1 description added where upstream had none; the organisation placeholder has been made dynamic; the elapsed-time scale follows CEM v3.1 revision 5 with seven options (AS41's estimate is labelled ≤ three months); the first stakeholder and the stakeholder names of 16 assumptions are the organisation placeholder. Restored as framework bindings: the attack-step mappings to the CRA (24 attack steps, 33 letters). The help texts of the personal-data questions QI2 and QI11 are filed under the option each explains (only their keys moved). The lowest risk level is renamed 'Negligible' (formerly 'No Risk').

Warranty: Unless required by applicable law or agreed to in writing, the licensor provides the work "as is", without warranties of any kind.

Cyber Resilience Act (Regulation (EU) 2024/2847): Annex I Part I, Annex I Part II, Annex II 1.0

EU law; free to reproduce.

The requirement texts are those of Regulation (EU) 2024/2847. A verdict here is a projection of this assessment, not a presumption of conformity.

ISA/IEC 62443-4-2 1.0

IEC copyright. Control ids and the requirement titles the packs cite only; no clause text.

Control ids and titles only; consult ISA/IEC 62443-4-2 for the requirements. A verdict here is a projection of this assessment, not a certification.

© Atrekos. All rights reserved.

Radio Equipment Directive 2014/53/EU, Article 3(3)(d), (e): EN 18031-1, -2 1.0

Control ids only: the requirement codes of EN 18031-1:2024 and EN 18031-2:2024 (for example ACM-1) and, in the restriction notes, the clause numbers the Official Journal names. None of the standards' text is reproduced. The restriction and validity wording is that of the Official Journal acts cited. No licence finding on reproducing the text exists; until one does, ids only.

CEN and CENELEC neither endorse nor have reviewed this mapping. A verdict here is a projection of this assessment that supports the manufacturer's technical documentation; it is not a presumption of conformity: the sections of each standard named "rationale" and "guidance" confer none (Implementing Decision (EU) 2025/138), and a control marked Restricted carries the further restriction named on it. Scope of this version: EN 18031-1 (Article 3(3)(d)) and EN 18031-2 (Article 3(3)(e)) for internet-connected radio equipment. EN 18031-3 (Article 3(3)(f), equipment that transfers money, monetary value or virtual currency) is out of scope; such equipment needs a separate assessment. Within EN 18031-2, only equipment processing personal data is judged; the toy, childcare and wearable clauses, and traffic and location data, are out of scope because the questionnaire does not ask about them. Applicability is decided by the answers: QA5 (wireless connections) with QA6 or QA10b (internet connection) for Part 1, and QI2 (personal data) in addition for Part 2. QA5 asks about wireless connections more broadly than radio equipment, so the product's radio role decides whether this framework applies. Each requirement shows the EN IEC 62443-4-2:2019 requirements its part's informative Annex B names; these links are shown and never change a verdict. It applies to radio equipment placed on the market between 1 August 2025 and 10 December 2027; Delegated Regulation (EU) 2022/30 is repealed with effect from 11 December 2027 (Delegated Regulation (EU) 2026/339).

© Atrekos. All rights reserved.